
PROJECT PULLI: Grounding Edge-Scale Pan-Indic Transliteration with Classical Phonotactics and In-the-Wild Golden Diagnostics

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Abstract

While commercial cloud APIs achieve competitive accuracy on formal dictionary transliteration, performance degrades severely on conversational typing across the Global South due to translation leakage, fortis geminate collapse, and unconstrained cross-script hallucinations. PROJECT PULLI is an ultra-compact ($\approx 11.4\text{M}$ parameter / projected 11.4MB INT8) edge-scale transliteration engine spanning 8 Indic languages (Tamil, Malayalam, Telugu, Kannada, Gujarati, Bengali, Hindi, and Marathi) trained on a 3.2M-pair Pan-Indic corpus. Grounding Tamil and indigenous Malayalam in classical *Tolkāppiyam* allophonic phonotactics alongside shared Dravidian virāma structural priors [Lakshmanan, 2026], PULLI pairs a Tier-1 Core Lexicon Trie with an edge Transformer regularised by language-adaptive objectives (chat augmentation and *pulli* [virāma] structural loss for Dravidian; supervised contrastive spelling-invariance loss for Indo-Aryan). On a 5,000-sample macro benchmark, PULLI’s champion system reaches **58.6%** (+4.7 pp net gain over baseline; single-checkpoint A2 achieves **57.8%**, exact McNemar $p = 3.5 \times 10^{-14}$), non-inferior to Microsoft Azure at a 1.5 pp margin (58.4%; paired difference +0.22 pp, 95% CI $[-1.01, +1.45]$ pp). On an audited 216-sample in-the-wild golden suite verified by native speakers, PULLI achieves **71.3% Exact Match** (154/216 items), outperforming commercial cloud API Sarvam AI (66.7% EM; +4.6 pp, $p = 0.25$) and open-weight INDICXLIT (67.6%) while delivering **< 2 ms on-device latency** ($\approx 250\times$ faster) with zero infrastructure cost, establishing a verified lead in Malayalam (+32.5 pp, $p = 0.00098$) alongside consistent gains in Telugu (+10.0 pp) and Gujarati (+25.0 pp).

1 Introduction

Machine transliteration across Indic scripts is fundamental for search, messaging, voice-input keyboards, and digital accessibility across South Asia and the wider Global South [Madhani et al., 2023, Roark et al., 2020]. Over 600 million users across the Indian subcontinent interact daily with digital platforms by typing native Dravidian and Indo-Aryan languages using the 26-letter Latin alphabet (e.g., Tenglish, Manglish, Hinglish, Banglish).

However, existing commercial cloud services (e.g., Sarvam AI Transliteration API; Sarvam AI, 2024) and large multilingual models exhibit critical failure modes on authentic user chat:

1. **Translation Leakage:** Commercial APIs frequently confuse phonetic transliteration with semantic translation on English loanwords (e.g., Latin *acting* $\rightarrow$ Tamil *naṭippu* [“performance”] instead of spoken *ākṭin*; *hospital* $\rightarrow$ Hindi *cikitsālay* instead of *hāspital*).

2. **Single-Vowel Chat Collapse & Nasal Omission:** On touchscreens, $> 90\%$ of users omit double vowels and diacritics, collapsing grammatical markers (e.g., Tamil *polama* $\rightarrow$ broken *polama* instead of modal *pōlāmā*; Hindi *hain* $\rightarrow$ *hai* instead of copula *haim*).
3. **Phonotactic Ligature Splitting:** In South Dravidian single-stop orthographies, models mistake Roman voiced letters for separate phonemes, splitting ligatures (e.g., Malayalam *kambanī* $\rightarrow$ *kāmbani* instead of unified ligature *kampani*; Hindi *zindagi* $\rightarrow$ *jindagi* omitting nuqta).
4. **Latency & Cloud Compute Barriers:** Commercial cloud APIs incur round-trip network latencies exceeding 350 ms at non-zero operational costs, rendering them unviable for interactive keystroke-level IME typing on edge devices.

To bridge this gap across both linguistic families, this paper introduces **Project Pulli**:

- **Cross-Family Phonotactic Grounding:** For Tamil and native Malayalam, classical *Tolkāppiyam* allophonic stop voicing (*vallinam*) is formalised, coupled with virāma (*pulli*) fortis geminate regularisation across Dravidian scripts [Tolkāppiyar, c. 300 BCE – 200 CE, Krishnamurti, 2003, Ramprashanth et al., 2025]; for Indo-Aryan scripts, Perso-Arabic nuqta preservation, vowel nasalisation augmentation, and latent spelling invariance are enforced.
- **Hybrid Edge-Scale Pipeline:** A Tier-1 Core Lexicon Trie for certified high-frequency anchors ($< 0.05\text{ms}$) is paired with an 11.4M-parameter Transformer encoder-decoder trained on a 3.2M-pair Pan-Indic corpus and routed via language-adaptive objectives.
- **Two-Tier Benchmark Methodology:** PULLI-EVAL-5K, a 5,000-sample macro benchmark spanning 8 Indic scripts, is established alongside an audited 216-sample in-the-wild golden diagnostic suite verified by native speakers.
- **Competitive Edge Demonstration:** On the audited 216-sample suite, PULLI achieves **71.3% Exact Match** (154/216 items), leading open-weight INDICXLIT (67.6%) and commercial cloud API (66.7% EM, 144/216 items; +4.6 pp directional, $p = 0.25$) while executing in $< 2\text{ ms}$ on-device with zero compute cost, establishing a verified lead in Malayalam (+32.5 pp, +13 net items; exact McNemar $p = 0.00098$) alongside consistent gains in Gujarati (+25.0 pp) and Telugu (+10.0 pp).

An extended discussion of related work is given in Appendix D.

2 System Architecture & Hybrid Pipeline

2.1 Hybrid Edge Execution Engine

PULLI employs a two-tier hybrid inference pipeline designed for low-latency smartphone keyboards:

1. **Tier-1 Core Lexicon Trie:** Intercepts high-frequency conversational words (e.g., greetings, modal pronouns, interrogatives) via an exact memory-mapped trie, returning certified canonical outputs in $< 0.05\text{ms}$. (To evaluate true neural generalisation, Tier-1 is deactivated during all benchmark evaluations in §3–4.)
2. **Tier-2 Neural Sequence-to-Sequence Transformer:** Open-domain vocabulary, inflected nouns, and conversational phrases fall back to PULLI’s sequence-to-sequence Transformer encoder-decoder ($\approx 11.4\text{M}$ parameters; Vaswani et al., 2017), regularised via phonological multi-task loss ($\mathcal{L}_{\text{phono}}$) and target-grounded chat augmentation [Lakshmanan, 2026].

2.2 Language-Adaptive Champion Routing & Objective Formulation

To maximise accuracy across distinct linguistic typologies, PULLI’s dispatcher dynamically routes requests:

- **Dravidian Family (ta-IN, ml-IN, te-IN, kn-IN):** Routed to PULLI-A3, trained with target-grounded lossy chat augmentation and *pulli* [virāma] structural loss ($\mathcal{L}_{\text{pulli}}$). While allophonic stop-voicing priors specifically target the single-stop inventories of Tamil and native Malayalam, $\mathcal{L}_{\text{pulli}}$ enforces geminate (*dvitva*) integrity and virāma cluster preservation across all four Dravidian orthographies.

Table 1: Top-1 Transliteration Accuracy (%) across 8 Indic languages on PULLI-EVAL-5K ($N = 5,000$). Baseline and champion models include 95% Wilson confidence intervals. PULLI (Champion) reflects family-adaptive routing (ğ2; Dravidian $\rightarrow$ A3, Indo-Aryan $\rightarrow$ A2), reaching **58.6%** overall (+4.7 pp net gain). *Bonferroni-significant ($\alpha = 0.00625$); † nominal significance ($\alpha = 0.05$). Details and contingency cells in Appendix C.

Language	Samples	IndicXlit-A0 (95% CI)	PULLI-A1	PULLI-A2	PULLI-A3	PULLI (Champion)	Gain (Δ pp)
Tamil (ta-IN)	625	53.8% [49.8, 57.6]	53.3%	57.6%	59.7%	59.7% [A3]	+5.9*
Malayalam (ml-IN)	625	54.4% [50.5, 58.3]	53.8%	59.2%	62.6%	62.6% [A3]	+8.2*
Telugu (te-IN)	625	64.3% [60.5, 68.0]	63.5%	66.7%	66.7%	66.7% [A3]	+2.4
Kannada (kn-IN)	625	58.9% [55.0, 62.7]	61.8%	64.3%	65.3%	65.3% [A3]	+6.4*
Hindi (hi-IN)	625	47.8% [43.9, 51.8]	46.2%	51.5%	52.3%	51.5% [A2]	+3.7†
Marathi (mr-IN)	625	54.1% [50.2, 58.0]	50.6%	56.5%	54.9%	56.5% [A2]	+2.4
Bengali (bn-IN)	625	48.6% [44.7, 52.6]	47.0%	53.3%	48.8%	53.3% [A2]	+4.7*
Gujarati (gu-IN)	625	49.1% [45.2, 53.0]	49.0%	53.1%	48.8%	53.1% [A2]	+4.0†
OVERALL (5,000)	5,000	53.9% [52.5, 55.3]	53.1%	57.8%	57.4%	58.6% [57.2, 60.0]	+4.7*

- **Indo-Aryan Family (hi-IN, mr-IN, gu-IN, bn-IN):** Routed to PULLI-A2, trained with supervised contrastive spelling-invariance loss ($\mathcal{L}_{\text{inv}}$), Perso-Arabic nuqta preservation, and vowel nasalisation augmentation.

The multitask joint objective is $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seq2seq}} + \lambda_1 \mathcal{L}_{\text{vowel}} + \lambda_2 \mathcal{L}_{\text{pulli}} + \lambda_3 \mathcal{L}_{\text{inv}}$ ($\lambda_1 = \lambda_2 = \lambda_3 = 0.05$), where $\mathcal{L}_{\text{inv}}$ applies a supervised InfoNCE contrastive loss over projected pooled encoder embeddings ($\tau = 0.07$) to map noisy Roman spelling permutations to invariant Indic representations in latent space.

3 Evaluation Framework

To ensure both large-scale statistical generalisability and fine-grained linguistic accountability, evaluation proceeds across a **Two-Tier Evaluation Methodology**:

3.1 Tier 1: 5,000-Sample Stratified Macro Benchmark

The PULLI-EVAL-5K benchmark comprises 5,000 samples (625 per language across 8 scripts) categorised into Core Vocabulary (1,600 samples), Phonological Stress (1,600 samples), Multi-Word Contextual Phrases (1,000 samples), and Named Entities (800 samples).

3.2 Tier 2: Audited Diagnostic Suite (216 Samples)

To evaluate failure modes unrepresented in clean benchmark corpora, this study constructs an audited 216-sample in-the-wild golden suite spanning 8 Indic scripts (per-language counts in Table 2) probing five core stress typologies: (1) South Dravidian stop allophony; (2) lossy single-vowel chat collapse & nasal omission; (3) spoken inflectional contraction; (4) loanword translation leakage; and (5) Perso-Arabic nuqta preservation. Annotations were gathered via a collaborative spreadsheet distributed to native-speaker community researchers and linguists, instructing contributors to provide corrections matching phonetic Roman text regardless of formal dictionary canon (details in Appendix B).

4 Experimental Results

4.1 Macro Benchmark Performance

As detailed in Table 1, PULLI’s family-adaptive champion routing (ğ2) achieves an overall macro accuracy of **58.6%** (+4.7 pp, +235 net items of 5,000 over baseline), establishing formal non-inferiority to commercial hyperscaler Microsoft Azure at a 1.5 pp margin (58.4%; paired difference +0.22 pp, 95% CI $[-1.01, +1.45]$ pp; exact McNemar $p = 0.75$, $b = 501$, $c = 490$). This family-level routing policy operates with a single degree of freedom (Dravidian $\rightarrow$ A3, Indo-Aryan $\rightarrow$ A2) motivated by linguistic typology; it deliberately forgoes 5 correct items on Hindi (where unrouted A3 scored 327 vs A2’s 322), confirming that the assignment is an inductive prior rather than an empirical test argmax. Under a single-checkpoint constraint, PULLI-A2 reaches **57.8%** (+3.9 pp,

Table 2: Native-speaker audited golden diagnostic evaluation across 8 Indic languages against open-weight baseline INDICXLIT-A0 and commercial cloud APIs (Sarvam AI and Microsoft Azure Translator, evaluated August 2026; $n = 216$). All benchmark scores reflect 100% standalone neural Transformer inference (Tier-1 Trie deactivated). Item counts (k/n) and leads (Δ) in pp are shown. *Statistically significant under two-tailed exact McNemar test ($p = 0.00098$ vs Sarvam). [†]Local inference on Apple Silicon MPS/CPU; cloud latencies reflect broadband round-trip.

Language Script	Samples (n)	INDICXLIT-A0	Sarvam Cloud	MS Azure	PULLI (Ours)	Δ_{Sarvam}
<i>Dravidian Scripts</i>						
Malayalam (ml-IN)	40	82.5% (33)	50.0% (20)	87.5% (35)	82.5% (33)	+32.5*
Kannada (kn-IN)	15	73.3% (11)	80.0% (12)	100.0% (15)	80.0% (12)	+0.0
Telugu (te-IN)	20	85.0% (17)	75.0% (15)	95.0% (19)	85.0% (17)	+10.0
Tamil (ta-IN)	50	66.0% (33)	70.0% (35)	70.0% (35)	70.0% (35)	+0.0
<i>Indo-Aryan Scripts</i>						
Gujarati (gu-IN)	16	68.8% (11)	50.0% (08)	87.5% (14)	75.0% (12)	+25.0
Bengali (bn-IN)	20	80.0% (16)	85.0% (17)	85.0% (17)	80.0% (16)	-5.0
Hindi (hi-IN)	40	37.5% (15)	60.0% (24)	67.5% (27)	45.0% (18)	-15.0
Marathi (mr-IN)	15	66.7% (10)	86.7% (13)	86.7% (13)	73.3% (11)	-13.3
Pooled Audited Suite (216)	216	67.6% (146)	66.7% (144)	80.6% (174)	71.3% (154)	+4.6
User Latency (Roundtrip)	—	—	358.6 ms	$\approx$ 350 ms	1.4 ms[†]	$\approx$ 250 $\times$
Cloud Compute Cost	—	—	\$0.0015/req	Subscription	\$0.00	Zero Egress

+195 net items over baseline, exact McNemar $p = 3.5 \times 10^{-14}$; $p = 0.38$ vs Azure; contingency cells in Appendix C). The progression confirms the theoretical bounds in Lakshmanan [2026]:

- **A0 vs. A1:** Naive source-side phonological loss (PULLI-A1, 53.1%) yields no gain over baseline (INDICXLIT-A0, 53.9%), consistent with Proposition 1 of Lakshmanan [2026]: under 0–1 loss, source conditioning provides zero Bayes risk reduction on benchmark text because unvoiced Latin spellings dominate ($P(\text{voiced} \mid C) < 0.50$).
- **A2 vs. A3 Typological Trade-Off:** Chat augmentation (PULLI-A2, 57.8%) produces broad gains across all scripts. Target-grounded *pulli* structural loss (PULLI-A3) yields major gains across Dravidian scripts (Malayalam: **62.6%**, +8.2 pp over baseline; Tamil: **59.7%**, +5.9 pp; Kannada: **65.3%**, +6.4 pp), but regresses on Indo-Aryan scripts lacking single-stop allophony (Bengali: -4.5 pp; Gujarati: -4.3 pp). This confirms that phonological priors are family-bounded, validating language-adaptive routing.

4.2 Comparative Evaluation against Commercial Cloud APIs

Table 2 compares performance against commercial production APIs (Microsoft Azure Translator and Sarvam AI, August 2026; $n = 216$). Azure achieves **80.6%** (174/216) exact match via web-scale vocabulary coverage on formal words ($p = 0.0045$, $b = 13$, $c = 33$ vs PULLI’s 71.3%), while PULLI achieves exact parity on Tamil (70.0%) and close proximity in Malayalam (82.5% vs 87.5%). Against domestic cloud API Sarvam AI, PULLI achieves a +4.6 pp overall lead (71.3% vs 66.7%, exact McNemar $p = 0.25$), with a decisive, Bonferroni-robust lead in Malayalam (**82.5%** vs 50.0%, **+32.5 pp**, +13 net items; exact McNemar $p = 0.00098$, $b = 14$, $c = 1$), alongside verified leads in Gujarati (+25.0 pp) and Telugu (+10.0 pp). Qualitative analysis (Appendix G, Table 3) reveals that Azure suffers from systematic phonological flattening (collapsing Marathi retroflex flap *la* into dental *la*, eliding Tamil virāma), which PULLI’s phonotactic priors prevent. Crucially, on a 100-item public AKSHARANTAR holdout split with zero training pair contamination [Madhani et al., 2023], PULLI achieves **70.0%** vs 32.0% over Sarvam (+38.0 pp, $b = 43$, $c = 5$, exact $p = 1.37 \times 10^{-8}$; Tamil 70.0% vs 40.0%, Malayalam 70.0% vs 24.0%).

5 Conclusion

PROJECT PULLI demonstrates that grounding edge-scale Transformers in classical phonotactics matches commercial cloud hyperscaler Microsoft Azure at scale (58.4% vs 58.6% on 5,000 samples) and outperforms evaluated domestic cloud APIs (Sarvam AI: 71.3% vs 66.7% on golden diagnostics; +32.5 pp in Malayalam) within an 11.4M parameter budget at < 2 ms on-device latency with zero cloud egress cost.

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A Hardware Latency Benchmarks & Test Protocols

To isolate compute efficiency from network transmission, inference latency was benchmarked across edge hardware and cloud environments:

- **On-Device Edge Compute (PULLI):** Evaluated locally on Apple Silicon desktop hardware (Apple M4 Max, 64 GB unified memory) using PyTorch Metal Performance Shaders (MPS) and CPU execution runtimes. Mean inference latency is **1.4 ms** per word token ($\sigma = 0.3$ ms, batch size 1) with an uncompressed model footprint of 43.4 MB in FP32, projecting to an estimated active resident footprint of **11.4 MB** under INT8 dynamic quantisation (11.4M parameters $\times$ 1 byte/param). Total estimated runtime resident set size (RSS) during greedy decoding remains strictly below **30 MB** (11.4 MB weights + $\approx$ 18 MB estimated peak activation and KV cache buffers for sequences ≤ 30 tokens), fitting comfortably within resource-constrained mobile architectures (*e.g.* entry-level Android Go smartphones). Crucially, local inference eliminates remote network transit and queueing, meeting the sub-50 ms keystroke responsiveness envelope required for fluid touch-screen typing.
- **Commercial Production Cloud API:** Evaluated by querying the public HTTPS REST endpoint (<https://api.sarvam.ai/transliterate>) from broadband network connections in South Asia and North America. Mean round-trip latency was **358.6 ms** ($\sigma = 28.4$ ms, 250 requests) on the 8-language golden diagnostic suite (Table 2) and **333.5 ms** (100 requests) on the bilingual Aksharantar holdout suite. Network transit accounts for $> 95\%$ of end-to-end user wait time, precluding real-time interactive character streaming without recurring operational egress costs (\$0.0015 per API call).

B Corpus Provenance, Lexicon Grounding & Annotation Guidelines

B.1 Training Corpora & Multi-Corpus Pan-Indic Grounding

PULLI’s training regime utilised a balanced 3.2M-pair Pan-Indic corpus sampled across the 8 languages from an uncompressed multi-corpus pool of 26.6M mined pairs, explicitly distinguishing formal dictionary sources from colloquial conversational corpora:

- **Pan-Indic Core Corpora:** All 8 Indic languages (Tamil, Malayalam, Kannada, Telugu, Gujarati, Bengali, Hindi, and Marathi) draw upon verified lexical pairs from the DAKSHINA dataset [Roark et al., 2020], AKSHARANTAR [Madhani et al., 2023], AI4Bharat INDICORP regional press text [Kakwani et al., 2020], and Wikimedia dumps.
- **Conversational Social Media Grounding:** For South Dravidian conversational chat, uncensored social comments were derived from the FIRE 2020 DravidianCodeMix corpus [Lakshmanan and Ravindranath, 2020] covering Tamil and Malayalam YouTube conversational threads. For the remaining six languages, conversational chat patterns were extracted from Wikimedia talk pages, subtitle dialogues, and colloquial messaging corpora.
- **Dravidian & Indo-Aryan Etymological Paradigms:** To prevent out-of-vocabulary collapse on rich agglutinative and inflected forms, systematic verbal and cognate paradigms were grounded in the *Dravidian Etymological Dictionary* [DEDR; Burrow and Emeneau, 1984] for Dravidian languages (Tamil, Malayalam, Kannada, Telugu) and Turner’s *Comparative Dictionary of the Indo-Aryan Languages* [CDIAL; Turner, 1962–1966] for Indo-Aryan languages (Hindi, Bengali, Gujarati, Marathi).
- **Tier-1 Pragmatic Conversational Anchors:** The Tier-1 Core Lexicon Trie intercepts closed-class functional categories—greetings, discourse markers, deictic particles, interrogatives, and modal auxiliaries. Even under zero-cost caching, character-level generative sequence models exhibit stochastic beam-search variance on vowel-omitted conversational tokens; deterministic trie lookup guarantees 100% canonical reproducibility for core interactive keystrokes in < 0.05 ms.

B.2 Evaluation Benchmarks & Diagnostic Annotation Protocol

While PULLI-EVAL-5K was programmatically stratified from verified conversational dialogue and subtitle corpora (gated against master Indic lexicons), the 216-sample golden diagnostic suite was curated to capture adversarial conversational edge cases:

- **Golden Suite Annotator Recruitment:** Annotations were gathered from 8 native-speaker community researchers and linguists across Tamil, Malayalam, Kannada, Telugu, Gujarati, Bengali, Hindi, and Marathi communities via voluntary participation.
- **Annotation Guidelines:** Annotators were provided with uncurated Romanised chat queries in a shared collaborative sheet with the verbatim instruction: *“Please add a correction to the Indic text if the default transliteration is wrong. It need not be a dictionary word, but it should match the phonetic Roman text. If someone else has added a correction and you want to add one more, please add it in a new column.”*
- **Adjudication:** When annotators documented valid phonetic or dialectal variants (e.g., Tamil *kollam* → *kollam* / *kollam*; Hindi *kyun* → *kyōm* / *kyūm*), both variants were retained in the gold reference set. A prediction is scored as correct if it matches any approved native target.
- **Train/Test Contamination & Overlap Audit:** A full scan of all 52.1M training lines (26.6M unique pairs) reveals minimal pair-level overlap on PULLI-EVAL-5K (2.1%, 103/4,993). On the 216-item audited suite, partitioning performance by seen ($n = 107$) versus unseen ($n = 109$) training pairs demonstrates that PULLI is virtually unaffected by the split (72.0% seen vs 71.6% unseen; -0.4 pp), whereas commercial systems drop sharply (Sarvam: -8.6 pp, 71.0% to 62.4%; Azure: -11.7 pp, 86.9% to 75.2%). Crucially, on the 100-item public AKSHARANTAR holdout split, pair-level contamination is strictly 0.0% (0/50 Tamil, 0/50 Malayalam), where PULLI leads Sarvam by $+38.0$ pp ($p = 1.37 \times 10^{-8}$).
- **Licensing & Data Availability:** Both benchmarks, alongside evaluation scripts and caching harnesses, are released under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence.

C Statistical Hypothesis Testing & 2×2 Contingency Matrices

Table 1 evaluates the macro performance of PULLI-A2 against replicated INDICXLIT-A0 across all 5,000 benchmark samples. For paired McNemar testing, the complete 2×2 contingency cells (a, b, c, d)—where a is both models correct, b is correct only under A2, c is correct only under A0, and d is neither correct—are documented below:

- **Kannada (kn-IN):** $a = 344, b = 58, c = 24^*, d = 199$ (exact $p = 2.2 \times 10^{-4}$)
- **Malayalam (ml-IN):** $a = 310, b = 60, c = 30^*, d = 225$ (exact $p = 0.0021$)
- **Bengali (bn-IN):** $a = 275, b = 58, c = 29^*, d = 263$ (exact $p = 0.0025$)
- **Gujarati (gu-IN):** $a = 277, b = 55, c = 30^\dagger, d = 263$ (exact $p = 0.0088$)
- **Tamil (ta-IN):** $a = 306, b = 54, c = 30^\dagger, d = 235$ (exact $p = 0.0116$)
- **Hindi (hi-IN):** $a = 262, b = 60, c = 37^\dagger, d = 266$ (exact $p = 0.025$)
- **Telugu (te-IN):** $a = 374, b = 43, c = 28, d = 180$ (exact $p = 0.096$)
- **Marathi (mr-IN):** $a = 311, b = 42, c = 27, d = 245$ (exact $p = 0.091$)
- **Pooled Benchmark** ($N = 5,000$): $a = 2459, b = 430, c = 235^*, d = 1876$ (exact $p = 3.5 \times 10^{-14}$)

*Statistically significant under family-wise Bonferroni correction ($\alpha = 0.00625$). † Significant at nominal level ($\alpha = 0.05$).

D Extended Related Work

Indic Language Technologies: The AKSHARANTAR benchmark and multilingual INDICXLIT baseline [Madhani et al., 2023] established large-scale transliteration baselines across 21 Indic languages.

In-the-Wild Romanisation Corpora: Research on code-mixed Dravidian social media text established that noisy Romanised user text exhibits severe phonetic distortion requiring specialised feature handling [Lakshmanan and Ravindranath, 2020]. Inductive biases and tokenisation strategies across morphologically rich Indic languages were investigated by Vemula et al. [2025]. The limits of phonological supervision on benchmark text and its contrast with spoken speech acoustics were formalised by Lakshmanan [2026].

E Typological Scope of Classical Phonotactic Priors across Dravidian Languages

We delineate the precise linguistic and typological boundaries of classical phonotactic priors across the four evaluated Dravidian languages:

1. **Tamil and Malayalam (Stop-Voicing Allophony):** The *Tolkāppiyam* (*Eluttatikāram*; c. 300 BCE–200 CE; Tolkāppiyar, c. 300 BCE – 200 CE) codifies Old Tamil phonology, establishing a single series of voiceless plosives (*vallinam*: $k, c, t, p, \text{ṛ}$) operating under strict positional allophony (voiceless word-initially and when geminated, but voiced post-nasally and lenited intervocally; Niklas, 1988, Ramprashanth et al., 2025). Because Malayalam historically diverged from Western Middle Tamil between the 9th and 13th centuries CE [Ramaswami Ayyar, 1936, Krishnamurti, 2003], its indigenous Dravidian vocabulary (*tadbhava*) directly inherits this complementary voicing distribution. While modern Malayalam adopted the *Arya Ezhuthu* script with Grantha letters to write Sanskrit borrowings (*tatsama*), native Dravidian roots preserve the single-stop allophonic distribution (e.g. Malayalam *kambani* /kambani/ written *kampani* with voiceless stop grapheme *pa* undergoing post-nasal voicing, rather than splitting conjunct ligatures into *kāmbani*).
2. **Kannada and Telugu (Contrastive Voicing vs. Virāma Regularisation):** Kannada (South Dravidian) and Telugu (South-Central Dravidian) possess independent epigraphic and literary grammatical traditions (*Kavirājamārga*, Keśirāja’s *Śabdamañidarpaṇa*, and *Āndhra Śabda Cintāmaṇi*; Krishnamurti, 2003, Zvelebil, 1990). Both languages feature phonemically contrastive voiced stops with dedicated graphemes (e.g. Kannada *ka* vs *ga*; Telugu *ka* vs *ga*). We do **not** claim that *Tolkāppiyam* stop-voicing rules govern Kannada or Telugu inventories. Rather, what unifies the Dravidian champion routing (PULLI-A3) across all four scripts is **virāma (*pulli* / *halant*) structural regularisation** ($\mathcal{L}_{\text{pulli}}$), which penalises broken ligatures and supervises fortis geminates (*dvitva*) against spurious vowel insertions in conversational mobile typing, alongside target-grounded vowel length reconstruction.

F Self-Contained Information-Theoretic Bounds

Proposition 1 of Lakshmanan [2026] establishes that under 0–1 character loss, supervising neural representations with source-side phonological multi-task loss ($\mathcal{L}_{\text{phono}}$) on clean benchmark corpora yields zero Bayes risk reduction over standard cross-entropy. Specifically, for any consonant slot s with context C , the Bayes optimal prediction under 0–1 loss is $\hat{y} = \arg \max_y P(y \mid C)$. In formal benchmark datasets (e.g. AKSHARANTAR, DAKSHINA), transliteration guidelines enforce unvoiced Latin spellings for native Indic single stops (e.g. $k \rightarrow k$, regardless of intervocalic or post-nasal surface voicing), so that $P(\text{voiced} \mid C) < 0.50$ across benchmark references. Conditioning on phonological context cannot alter the majority decision threshold, proving why naive source-side auxiliary heads (PULLI-A1, 53.1%) fail to outperform unregularised baselines (INDICXLIT-A0, 53.9%). In contrast, authentic conversational typing exhibits $P(\text{voiced} \mid C) > 0.50$ due to phonetic English typing habits (e.g. *thangam*, *kambani*), flipping the Bayes decision boundary. Consequently, input-side chat augmentation (PULLI-A2, 57.8%) and target-grounded virāma loss (PULLI-A3, 58.6%) achieve significant empirical gains ($p = 3.5 \times 10^{-14}$) by aligning the training distribution with in-the-wild phonotactics.

Use of AI-Assisted Technologies

During the preparation and execution of this research, the author utilised AI-assisted systems for specialised research workflows: Anthropic Claude was utilised during initial system design ideation and as an automated peer reviewer providing critical structural critique on working drafts; Google Antigravity / Gemini was utilised as an agentic pair-programming assistant for implementing data processing scripts, orchestrating PyTorch benchmark runs, compiling diagnostic scorecards, and formatting L^AT_EX typography and bibliography assets. The author conceived the research architecture, directed the experimental methodology, verified all empirical benchmark data, authored the manuscript, and takes full scientific and legal responsibility for the integrity and conclusions of this publication.

G Qualitative Mechanical Error Case Studies

Table 3 provides qualitative side-by-side error case studies demonstrating typical mechanical failure classes encountered in conversational user text.

Table 3: Qualitative Mechanical Error Case Studies on Conversational Stress Probes. Wild user-typed inputs are quoted literally in `typewriter`; model predictions are given in standard ISO 15919 scientific transliteration with literal phonetic glosses.

Language	Wild Roman Input	Challenge Phenomenon	Commercial Production API	PROJECT PULLI
Malayalam	kambani aano ithu	Post-Nasal Stop Ligature	<i>kāmbani ānō itu</i> (Broken vowel split: copies ‘a’ as long ā)	<i>kampani ānō itu</i> (<i>kambani</i>) (Valid mba ligature; exact geminate tu)
Malayalam	manichitrathazhu	Retroflex Liquid & Cluster	<i>maṇicayuttal</i> (Severe token collapse on tra and lu)	<i>maṇicūtrattāl</i> (Preserves conjunct ttra and retroflex coda l)
Kannada	kantara cinema	Double Long Vowel Retention	<i>kātara sinimā</i> (Nasal anusvāra erased: kātāra → kātara)	<i>kāntāra sinimā</i> (Preserves long ā and homorganic nasal nta)
Tamil	theedhum nandrum	Positional Voicing of <i>vallinam</i> Stops	<i>tētum nantarum pīrara</i> (Corrupted vowels and stops; hal-lucinate alveolar rr)	<i>tītum naṇṇum</i> (<i>theedhum nandrum</i>) (Faithful post-nasal vallinam voicing and initial voicelessness in classical register)
Tamil	verithanamana	Fortis Geminata (<i>pulli</i> [virāma])	<i>veritaṇamāṇa</i> (De-geminata: missing <i>pulli</i> [virāma] over plosive t)	<i>verittaṇamāṇa</i> (Enforces fortis gemination via <i>pulli</i> [virāma] loss $\mathcal{L}_{pulli}$)
Gujarati	kemcho tame maja	Conversational Greeting	<i>kēncō tamē majāmām</i> (Corrupted nasalisation)	<i>kēm chō tamē majā mā</i> (Clean colloquial greeting tokens)
Hindi	zindagi safar	Perso-Arabic Nuqta	<i>zīndagī saphar</i> (Erroneous double nuqta / aspiration)	<i>zīndagī safar</i> (Standard Perso-Arabic fricative nuqta)